

Mehdi Dadfarnia

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Education

University of Maryland, College Park. *College Park, MD.*

Ph.D. Mechanical Engineering, 2025.

Advisor: Jeffrey W. Herrmann. Thesis: *Approaches to Evaluate Condition Monitoring-Based Technologies for Manufacturing Maintenance and Risk Management.*

University of Maryland, College Park. *College Park, MD.*

M.S. Systems Engineering, 2014.

Advisor: John S. Baras. Thesis: *Energy Harvesting Microgenerators for Body Sensor Networks.*

University of Maryland, College Park. *College Park, MD.*

B.S. Electrical Engineering, 2012.

Employment

Research Engineer, Communications Technology Laboratory, National Institute of Standards & Technology.
October 2021 - current

Research Engineer, Engineering Laboratory, National Institute of Standards & Technology.
March 2015 - October 2021

Engineer Trainee, Information Technology Laboratory, National Institute of Standards & Technology.
January 2013 - March 2015

Graduate Research Assistant, Institute for Systems Research, University of Maryland, College Park.
January 2013 - May 2014

Engineer Trainee, Engineering Laboratory, National Institute of Standards & Technology.
June 2011 - January 2013

Honors

United States Department of Commerce Bronze Medal Award; awarded for developing an open-source manufacturing simulator that enables users to plan production and compare AI-based and traditional maintenance policies, January 2025.

Institute for Systems Research's Outstanding Systems Engineering Undergraduate Award, University of Maryland; awarded for contributions to requirements analysis modeling that enabled comparative study of ambulance safety standards, December 2013.

Publications

Peer-reviewed journal articles

Peer-reviewed conference articles

Technical reports

Technical presentations and posters (excluding conference presentations)

Supervision

Professional activities

Conference organization

Program committees

Journal reviews

Conference reviews

Service

Institutional service

University departmental service

Skills

Engineering expertise: Discrete-event modeling & simulation, digital twin design, prognostics & health management, applied machine learning, risk analysis, engineering decision-making.

Engineering software: Simulink, SysML (proficient), Modelica, Arena (past experience).

Programming: Python, MATLAB (proficient), Julia, C/C++ (past experience).

Libraries: pandas, NumPy, SciPy, scikit-learn, Matplotlib, SimPy, ManPy (proficient).

Languages: English (native), Persian (proficient), Spanish (beginner).